Primary Keys

```
postgres=# CREATE TABLE students_id (
postgres(# student_id SERIAL PRIMARY KEY,
postgres(# name VARCHAR(100) NOT NULL,
postgres(# email VARCHAR(150) UNIQUE,
postgres(# faculty VARCHAR(50)
postgres(# );
CREATE TABLE
postgres=# INSERT INTO students_id (name, email, faculty) VALUES
postgres-# ('Alice Johnson', 'alice@alatoo.edu.kg', 'Computer Science'),
postgres-# ('Bob Smith', 'bob@alatoo.edu.kg', 'Mathematics'),
postgres-# ('Carol Davis', 'carol@alatoo.edu.kg', 'Computer Science');
INSERT 0 3
postgres=# CREATE TABLE course_enrollments (
postgres(# student_id INTEGER,
postgres(# course_id INTEGER,
postgres(# semester VARCHAR(20),
postgres(# enrollment_date DATE DEFAULT CURRENT_DATE,
postgres(# grade CHAR(2),
postgres(# PRIMARY KEY (student_id, course_id, semester)
postgres(# );
CREATE TABLE
postgres=# INSERT INTO course_enrollments (student_id, course_id, semester, grade)
postgres-# VALUES
postgres-# (1, 101, '2024-Spring', 'A'),
postgres-# (1, 102, '2024-Spring', 'B+'),
postgres-# (2, 101, '2024-Spring', 'A-'),
postgres-# (1, 101, '2024-Fall', 'A+');
INSERT 0 4
postgres=# SELECT * FROM students_id;
 student_id |     name      |        email        |     faculty
------------+---------------+---------------------+-----------------
          1 | Alice Johnson | alice@alatoo.edu.kg | Computer Science
          2 | Bob Smith     | bob@alatoo.edu.kg   | Mathematics
          3 | Carol Davis   | carol@alatoo.edu.kg | Computer Science
(3 rows)
```

```
postgres=# \c mydb
You are now connected to database "mydb" as user "postgres".
mydb=# SELECT * FROM mydb;
ERROR:  relation "mydb" does not exist
LINE 1: SELECT * FROM mydb;
                      ^
mydb=# SELECT * FROM users;
 id |  name   |         email         | age | registered_at | is_active | balance
----+---------+-----------------------+-----+---------------+-----------+---------
  1 | Alina   | gh45278@gmail.com     |  22 | 2024-05-12    | t         |  250.00
  2 | Emir    | ghdh4893@gmail.com    |  27 | 2025-02-18    | f         |    0.00
  3 | Yerlan  | usg@gmail.com         |  35 | 2023-11-30    | t         | 1420.50
  4 | Aidana  | aidana.k@gmail.com    |  19 | 2026-01-10    | t         |   15.75
  5 | Bekzat  | bekzat.t@gmail.com    |  31 | 2024-08-22    | t         |  890.00
  6 | Nurzat  | nurzat88@gmail.com    |  24 | 2025-07-04    | f         |    5.00
  7 | Aibek   | aibek.n@gmail.com     |  45 | 2022-03-15    | t         | 5300.00
  8 | Dinara  | dinara.zh@gmail.com   |  28 | 2025-10-01    | t         |  120.40
  9 | Kanat   | kanat.b@gmail.com     |  23 | 2025-12-25    | t         |    0.00
 10 | Meerim  | meerim.a@gmail.com    |  38 | 2023-04-19    | t         | 2150.80
 11 | Sultan  | sultan.k@gmail.com    |  28 | 2025-01-15    | t         |  150.50
 12 | Elena   | elena.v@mail.ru       |  34 | 2024-11-02    | t         | 1200.00
 13 | Timur   | timur99@gmail.com     |  21 | 2026-02-10    | f         |    0.00
 14 | Aigerim | aigerim.m@outlook.com |  26 | 2025-05-20    | t         |  450.75
 15 | Marat   | marat.b@gmail.com     |  42 | 2023-08-14    | t         |   89.20
 16 | Anna    | anna.smith@gmail.com  |  19 | 2026-01-01    | t         |   15.00
 17 | Ruslan  | ruslan.t@gmail.com    |  31 | 2024-04-25    | f         |  550.00
 18 | Klara   | klara.k@yandex.ru     |  50 | 2022-12-12    | t         | 3400.60
 19 | Danial  | danial.z@gmail.com    |  24 | 2025-09-05    | t         |    0.50
 20 | Madina  | madina.a@gmail.com    |  29 | 2025-03-18    | t         |  720.00
 21 | Sultan  | sultan.k@gmail.com    |  28 | 2025-01-15    | t         |  150.50
 22 | Elena   | elena.v@mail.ru       |  34 | 2024-11-02    | t         | 1200.00
 23 | Timur   | timur99@gmail.com     |  21 | 2026-02-10    | f         |    0.00
 24 | Aigerim | aigerim.m@outlook.com |  26 | 2025-05-20    | t         |  450.75
 25 | Marat   | marat.b@gmail.com     |  42 | 2023-08-14    | t         |   89.20
 26 | Anna    | anna.smith@gmail.com  |  19 | 2026-01-01    | t         |   15.00
```

```
mydb=*# \d users
                                  Table "public.users"
    Column     |          Type          | Collation | Nullable |              Default
---------------+------------------------+-----------+----------+-----------------------------------
 id            | integer                |           | not null | nextval('users_id_seq'::regclass)
 name          | character varying(50)  |           |          |
 email         | character varying(100) |           |          |
 age           | integer                |           |          |
 registered_at | date                   |           |          |
 is_active     | boolean                |           |          | true
 balance       | numeric(10,2)          |           |          | 0.00
Indexes:
    "users_pkey" PRIMARY KEY, btree (id)
    "users_email_unique" UNIQUE CONSTRAINT, btree (email)
```

```
postgres=# CREATE TABLE transactions (
postgres(#     transaction_id BIGSERIAL PRIMARY KEY,
postgres(#     amount DECIMAL(15, 2),
postgres(#     transaction_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP
postgres(# );
CREATE TABLE
postgres=# INSERT INTO transactions (amount, transaction_date) VALUES
postgres-#     (1500.00,  '2026-09-01 10:15:00'),
postgres-#     (249.99,   '2026-09-03 14:42:10'),
postgres-#     (87500.50, '2026-09-10 09:05:33'),
postgres-#     (12.75,    '2026-09-15 18:20:00'),
postgres-#     (3200.00,  '2026-10-01 11:30:45');
INSERT 0 5
postgres=# SELECT * FROM transactions;
 transaction_id |  amount   |  transaction_date
----------------+-----------+---------------------
              1 |  1500.00 | 2026-09-01 10:15:00
              2 |   249.99 | 2026-09-03 14:42:10
              3 | 87500.50 | 2026-09-10 09:05:33
              4 |    12.75 | 2026-09-15 18:20:00
              5 |  3200.00 | 2026-10-01 11:30:45
(5 rows)
```

```
mydb=# CREATE TABLE orders (
mydb(#     order_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
mydb(#     customer_name VARCHAR(100),
mydb(#     order_date DATE DEFAULT CURRENT_DATE
mydb(# );
CREATE TABLE
mydb=#
mydb=# INSERT INTO orders (customer_name, order_date) VALUES
mydb-#     ('Alina',   '2026-08-01'),
mydb-#     ('Emir',    '2026-08-03'),
mydb-#     ('Yerlan',  '2026-08-07'),
mydb-#     ('Aidana',  '2026-08-12'),
mydb-#     ('Bekzat',  '2026-08-15'),
mydb-#     ('Nurzat',  '2026-08-19'),
mydb-#     ('Aibek',   '2026-08-23'),
mydb-#     ('Dinara',  '2026-08-27'),
mydb-#     ('Kanat',   '2026-09-01'),
mydb-#     ('Meerim',  '2026-09-05'),
mydb-#     ('Sultan',  '2026-09-09'),
mydb-#     ('Elena',   '2026-09-14'),
mydb-#     ('Timur',   '2026-09-18'),
mydb-#     ('Madina',  '2026-09-25'),
mydb-#     ('Ruslan',  '2026-10-02');
INSERT 0 15
mydb=# SELECT * FROM orders;
 order_id | customer_name | order_date
----------+---------------+------------
        1 | Alina         | 2026-08-01
        2 | Emir          | 2026-08-03
        3 | Yerlan        | 2026-08-07
        4 | Aidana        | 2026-08-12
        5 | Bekzat        | 2026-08-15
        6 | Nurzat        | 2026-08-19
        7 | Aibek         | 2026-08-23
        8 | Dinara        | 2026-08-27
```

```
mydb=# \d orders
                                  Table "public.orders"
    Column     |          Type           | Collation | Nullable |           Default
---------------+-------------------------+-----------+----------+------------------------------
 order_id      | integer                 |           | not null | generated always as identity
 customer_name | character varying(100) |           |          |
 order_date    | date                    |           |          | CURRENT_DATE
Indexes:
    "orders_pkey" PRIMARY KEY, btree (order_id)
```